

High School

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1 Reasoning with Real Numbers

1.1 Exercise

Prove that $2 + 2 = 5$.

Solution

$$\begin{aligned}0 &= 0 \rightarrow 20 - 16 - 4 = 25 - 20 - 5 \\4(5 - 4 - 1) &= 5(5 - 4 - 1) \\4 &= 5 \\2 + 2 &= 5 \quad \blacksquare\end{aligned}$$

↑ This is a joke, not a real proof.

And that's because the number $5 - 4 - 1$ is equal to 0, and we cannot divide by 0.

2 Logarithms

2.1 Exercise

Convert the following exponential equation $40^{x-1} = 2^{2x+1}$ into logarithmic form.

Solution

$$\begin{aligned}40^x \cdot \frac{1}{40} &= 2^{2x} \cdot 2 \\40^x &= 2^{2x} \cdot 80 \\40^x &= 4^x \cdot 80 \\10^x &= 80 \\\log_{10} 80 &= x\end{aligned}$$

2.2 Exercise

Prove that if $x > 1$, then:

$$2 \log_2 (x^2 - 1) - 3 \log_2 (x - 1) = \log_2 \frac{x^2 + 2x + 1}{x - 1}$$

Solution

$$D : x > 1$$

$$L = 2 \log_2 (x^2 - 1) - 3 \log_2 (x - 1)$$

$$R = \log_2 \frac{x^2 + 2x + 1}{x - 1}$$

$$\begin{aligned} L &= \log_2 (x^2 - 1)^2 - \log_2 (x - 1)^3 = \log_2 \frac{(x^2 - 1)^2}{(x - 1)^3} = \log_2 \frac{\cancel{(x - 1)^2} (x + 1)^2}{\cancel{(x - 1)^2} (x - 1)} = \\ &= \log_2 \frac{x^2 + 2x + 1}{x - 1} = R \quad \blacksquare \end{aligned}$$

2.3 Exercise

Prove that $\log_{17} 19 : \log_{18} 19 = \log_{16} 18 \cdot \log_{17} 16$.

Solution

$$\log_{17} 19 \cdot \log_{19} 18 = \log_{17} 16 \cdot \log_{16} 18$$

$$\log_{17} \cancel{19} \cdot \log_{\cancel{19}} 18 = \log_{17} \cancel{16} \cdot \log_{\cancel{16}} 18$$

$$\log_{17} 18 = \log_{17} 18$$

$$L = R \quad \blacksquare$$

2.4 Exercise

Given the numbers $a = \log_3 2$ and $b = \log_{2^{2024}} 3^{1012} + \log_{\frac{1}{3}} 54$. Express b in terms of a .

Solution

$$\begin{aligned} b &= \frac{1}{2} \cdot \log_2 3 - \log_3 (27 \cdot 2) = \frac{1}{2} \cdot \frac{1}{a} - (\log_3 3^3 + a) = \frac{1}{2a} - 3 - a = \frac{1}{2a} - \frac{6a}{2a} - \frac{2a^2}{2a} = \\ &= -\frac{2a^2 + 6a - 1}{2a} \end{aligned}$$

2.5 Exercise

Given that $\frac{1}{\log_{p-q} 3} = 3$ and $(\log_{pq} 3)^{-1} = 2$. Prove that the number $\sqrt{p^2 - 1 + q^2}$ is rational.

Solution

$$\log_2 p - q = 3 \quad \wedge \quad \log_3 pq = 2 \rightarrow pq = 3^2 = 9$$

$$p - q = 2^3 / 2$$

$$(p - q)^2 = 64$$

$$p^2 + q^2 - 2pq = 64$$

$$p^2 + q^2 = 82$$

$$\sqrt{p^2 - 1 + q^2} = \sqrt{p^2 + q^2 - 1} = \sqrt{82 - 1} = \sqrt{81} = 9 \in \mathbb{Q} \quad \blacksquare$$

2.6 Exercise

Given that $\log_a m = \sqrt{2}$. Evaluate $3 \log_{\sqrt{m}} am^2 - \log_{am} \frac{a^2}{m}$.

Solution

$$\begin{aligned} 3 \cdot 2 \log_m am^2 - \frac{\log_a \frac{a^2}{m}}{\log_a am} &= 6 \cdot \log_m a + 12 - \frac{\log_a a^2 - \log_a m}{\log_a a + \log_a m} = \\ &= \frac{6}{\sqrt{2}} + 12 - \frac{2 - \sqrt{2}}{1 + \sqrt{2}} = \frac{6\sqrt{2}}{2} + 12 - \frac{(2 - \sqrt{2})(1 - \sqrt{2})}{(1 + \sqrt{2})(1 - \sqrt{2})} = \\ &= \frac{24 + 6\sqrt{2}}{2} - \frac{2 - 2\sqrt{2} - \sqrt{2} + 2}{1 - 2} = \frac{24 + 6\sqrt{2}}{2} + 4 - 3\sqrt{2} = 16 \end{aligned}$$

2.7 Exercise

Prove that for $a, b, c > 1$ and $n \in \mathbb{N}$ the number $\frac{\log_a b^3 \cdot \log_b 2^c}{\log \sqrt[n]{a^n} \sqrt[4]{c}}$ is even.

Solution

$$\frac{\frac{3}{2} \log_a b \cdot \log_b c}{\frac{3}{4n} \log_a c} = \frac{\frac{3}{2} \log_a c}{\frac{3}{4n} \log_a c} = \frac{3}{2} \cdot \frac{4}{3} \cdot n = 2 \cdot n \quad \blacksquare$$